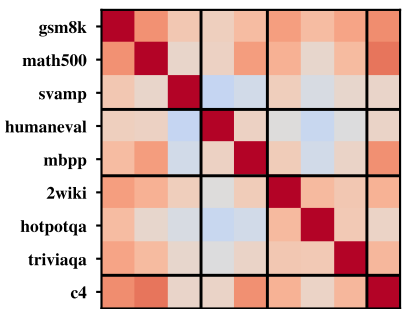
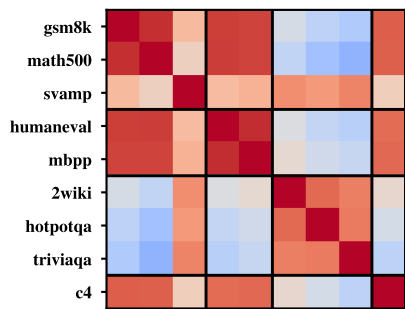
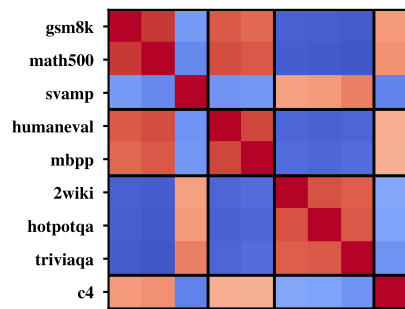
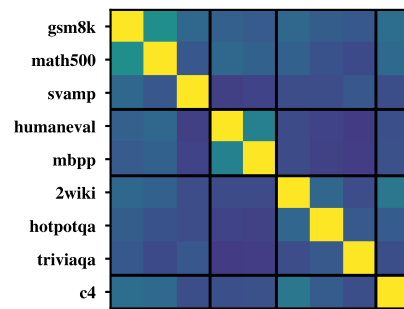


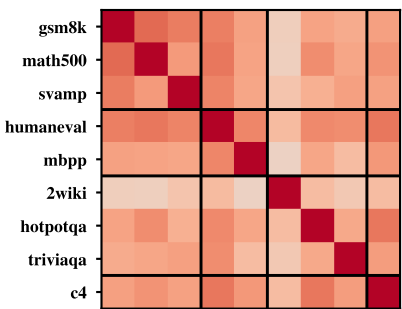
Raw Fisher cosine

**Log-Fisher cosine**

Residual log-Fisher cosine

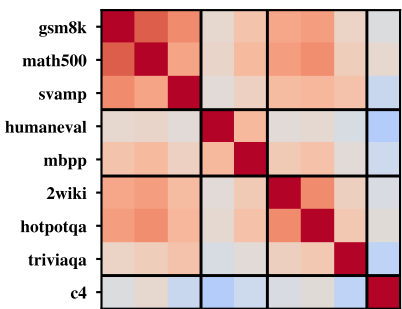
**Top-0.1% Jaccard**

Qwen--Qwen3-1.7B



Dataset	gsm8k	math500	svamp	humaneval	mbpp	2wiki	hotpotqa	triviaqa	c4
gsm8k	10	8	6	4	2	1	0	0	0
math500	8	10	7	5	3	2	1	0	0
svamp	6	7	10	8	6	4	2	1	0
humaneval	4	5	8	10	9	7	5	3	1
mbpp	2	3	6	9	10	8	6	4	2
2wiki	1	2	4	7	8	10	9	7	5
hotpotqa	0	1	2	5	6	9	10	8	6
triviaqa	0	0	1	3	4	7	8	10	9
c4	0	0	0	1	2	5	6	9	10

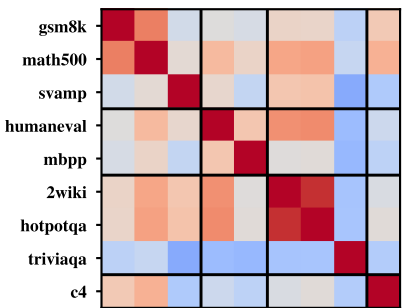
Qwen--Qwen3-4B

[illegible]

gsm8k									
math500									
svamp									
humaneval									
mbpp									
2wiki									
hotpotqa									
triviaqa									
c4									

Heatmap showing the performance of different models across various datasets. The datasets are gsm8k, math500, svamp, humaneval, mbpp, 2wiki, hotpotqa, triviaqa, and c4. The models are grouped into three categories: LLaMA (green), GPT (blue), and Gemini (red). The color scale ranges from 0.0 (dark) to 1.0 (light).

gemma3-1b

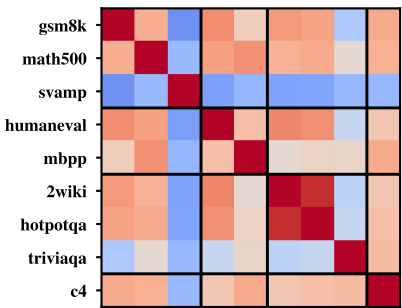


gsm8k							
math500							
svamp							
humaneval							
mbpp							
2wiki							
hotpotqa							
triviaqa							
c4							

gsm8k								
math500								
svamp								
humaneval								
mbpp								
2wiki								
hotpotqa								
triviaqa								
c4								

[illegible]

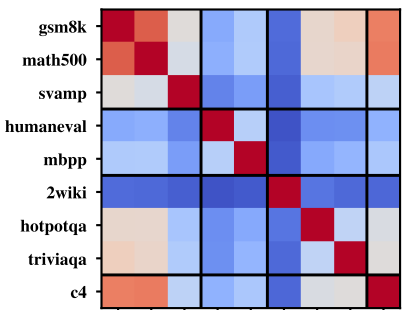
gemma3-270m

[illegible]

Heatmap showing the performance of different models across various datasets. The models are listed on the left: gsm8k, math500, svamp, humaneval, mbpp, 2wiki, hotpotqa, triviaqa, and c4. The datasets are listed on the top: gsm8k, math500, svamp, humaneval, mbpp, 2wiki, hotpotqa, triviaqa, and c4. The color scale ranges from 0.0 (dark blue) to 1.0 (dark red).

Dataset	Proposed Model (%)
gsm8k	~95
math500	~90
svamp	~95
humaneval	~90
mbpp	~95
2wiki	~90
hotpotqa	~95
triviaqa	~90
c4	~95

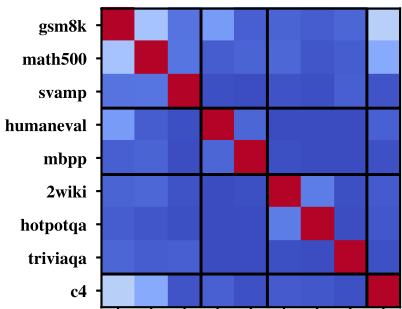
gemma3-4b

[illegible]

	gsm8k	math500	svamp	humaneval	mbpp	2wiki	hotpotqa	triviaqa	c4
gsm8k	Red	Red	Blue	Red	Red	Blue	Blue	Blue	Orange
math500	Red	Red	Blue	Red	Red	Blue	Blue	Blue	Orange
svamp	Blue	Blue	Red	Blue	Blue	Orange	Orange	Orange	Blue
humaneval	Red	Red	Blue	Red	Red	Blue	Blue	Blue	Orange
mbpp	Red	Red	Blue	Red	Red	Blue	Blue	Blue	Orange
2wiki	Blue	Blue	Orange	Blue	Blue	Red	Red	Red	Blue
hotpotqa	Blue	Blue	Orange	Blue	Blue	Red	Red	Red	Blue
triviaqa	Blue	Blue	Orange	Blue	Blue	Red	Red	Red	Blue
c4	Orange	Orange	Blue	Orange	Orange	Blue	Blue	Blue	Red

Heatmap showing the performance of different models across various datasets. The datasets are gsm8k, math500, svamp, humaneval, mbpp, 2wiki, hotpotqa, triviaqa, and c4. The models are grouped into three categories: LLMs (left), LLMs + CoT (middle), and LLMs + CoT + Self-Reflection (right). The color scale ranges from 0.0 (dark purple) to 1.0 (yellow).

olmo3-7b



gsm8k									
math500									
svamp									
humaneval									
mbpp									
2wiki									
hotpotqa									
triviaqa									
c4									

[illegible][illegible]